

# Tungnaá in Live Performance: An Implementation of Interactive Artistic Text-To-Voice

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## Abstract

We demonstrate Tungnaa, a software musical instrument for controlling interactive artistic text-to-voice (IATV) models. IATV is a proposed low-resource text-conditional voice generation task for musical performance applications. Tungnaa is a Python package implementing IATV training, a real-time inference engine and graphical interface for performance with IATV models. We also describe an application of Tungnaa in collaboration with sound poet Jaap Blonk.

**Index Terms:** Interactive Artistic Text-to-Voice, Controllable TTS, Artistic Exploration, Streaming Vocoder

## 1. Introduction

We demonstrate Tungnaa, a real-time software instrument that implements a proposed framework for interactive artistic text-to-voice (IATV). We also describe the process of creating a bespoke IATV dataset and Tungnaa model in collaboration with Dutch sound poet Jaap Blonk.

IATV is a new research task for voice synthesis situated within the field of interactive art. Much of the research landscape around artistic use of voice synthesis has focused on applications such as voice acting and pop vocal production. However, there are few options available for those with interests in abstract explorations of vocal sounds, interactive art, live performance, or improvisation. We see a desire to experiment with these kinds of systems within the artistic community, beginning with interest in the “WaveNet Aesthetic” of hyperrealistic and uncanny babbling first heard in a 2016 demo [1], and found in the work of contemporary musicians like Holly Herndon [2], DadaBots and Jennifer Walshe [3], the 2021 Mouse on Mars album *AAI*, and in the live performances of Jonathan Reus [4] and Kelsey Cotton [5].

These latter examples reflect the specific requirements of a voice system for live performance as embodied by work within the New Instruments for Musical Expression (NIME) research community [6], where an earlier iteration of this work was demonstrated [7].

## 2. Interactive Artistic Text-to-Voice

In a concurrent submission to the INTERSPEECH 2025 paper track, we propose IATV as a task and describe our efforts adapting TTS methods to solve it. Briefly, IATV is a text-conditional audio generation task with the following requirements:

1. Real-time performance with worst-case latency below 100 milliseconds, suitable for use in musical performance.
2. Interactivity, with human-in-the-loop manipulation of the synthesis process.

3. Controllability, exposing the underlying neural synthesis engine to induce alignment failures, artefacts and babbling.
4. Flexibility, without assumptions about voice style or method of text transcription (except for some temporal monotonic alignment between text and audio).
5. “Hi-Fi” audio, representing frequencies up to 20 kHz and with dynamic range suitable for digital music applications.
6. Openness, for users to train their own models, run them locally and integrate with other tools.
7. Customizability, with training on small datasets or bespoke text notations which may not resemble pretraining data.

Our proposed IATV system includes RAVE [8] as a streaming vocoder, which is to our knowledge its first use in a text-to-voice system. We use a pretrained token-free text encoder [9] to allow the creation of flexible text conditioning systems. Finally, we developed a Tacotron2-family [10] recurrent network for low-latency joint alignment and generation of vocoder parameters from small datasets.

## 3. Tungnaá

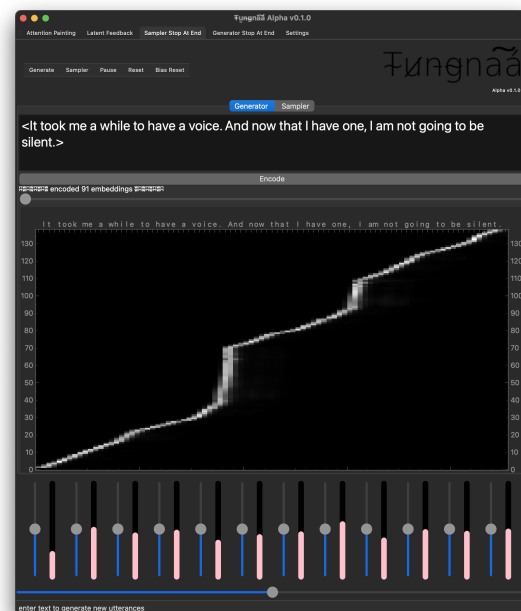


Figure 1: *The Tungnaá GUI.*

Tungnaa is a Python package<sup>1</sup> which incorporates training code for our IATV implementation, a real-time inference en-

<sup>1</sup><https://pypi.org/project/tungnaa>

gine, and a front-end musical instrument interface. A video demonstration is available in the accompanying materials.

### 3.1. Inference Engine

Our inference engine is built in Python using PyTorch and TorchScript [11]. The text encoder runs asynchronously in its own thread, computing new text features on demand as the user finishes entering a text. The recurrent IATV model and causal convolutional vocoder run in a different thread, which can buffer a few frames ahead to trade off stability for latency. Worker threads spend most time in C extensions, while the GUI runs in separate Python process, enabling low latency streaming inference without dropping audio frames. Tungnaá can run on a modern laptop under Linux and macOS.

### 3.2. Graphical Interface

The GUI is also built in Python using Qt. Functions of the GUI can also be controlled remotely using Open Sound Control [12] for integration with common computer music software.

Tungnaá's GUI consists of three areas: text entry, alignment, and vocoder (Figure 1). A text entry field allows the performer to provide text for Tungnaá to vocalise. The text can be sent on command to the text encoder, resetting the alignments.

A scrolling alignment graph depicts progress through the text over time. Once encoded, an input text appears along the horizontal axis of the graph, while time is on the vertical axis, with the present time at the top. Light pixels denote the portion of the text being used by the model at a given time. If *attention painting* mode is engaged, the *paint bar* allows the performer to directly manipulate this alignment. Otherwise, the system autonomously reads through the text according to its learned alignment model. By clicking into the alignment graph, a user can intervene by instantly moving the alignment (along the text axis) and return to a previous hidden state (along the time axis).

In the vocoder section, a set of meters and sliders display the RAVE latent vectors as they are produced by the alignment model, while a temperature control affects how variable they are. By manipulating the sliders, a performer can apply a bias to each latent variable in the vocoder, where dimensions related to loudness, voicedness, frontalization of vowels and so forth can often be discerned. With a *latent feedback* switch, the user chooses whether the biased vocoder signal is fed back into the alignment model to also affect the timing.

## 4. Application: Bla Blavatar vs Jaap Blonk

Tungnaá was used in a collaboration with the prolific Dutch sound poet Jaap Blonk to create the live performance *Bla Blavatar vs. Jaap Blonk*. Sound poetry is a form of poetry with origins in the European Futurist and Dada art movements which blurs the lines between poetry and music performance, often involving extensive exploration of voice sounds and the invention of new voice notation systems. As sound poetry is also meant to be experienced live, it serves as a suitable exemplar scenario for testing the efficacy of an IATV system.

In collaboration with Blonk, we created an IATV dataset consisting of approximately two hours of voice audio recordings paired with notations written in a heuristic phonetic system called RPA (Reduced Phonetic Alphabet). This dataset was used to train an IATV model which was then used with Tungnaá in live performance.

In each performance, Blonk performs a different algorithmically generated RPA score. Throughout the performance he

“battles” the Bla Blavatar, an artificial sound poet controlled by a second performer using Tungnaá. As part of the artwork, all of Blonk’s vocalisations are recorded and matched to the RPA score, expanding the dataset with each performance and contributing to a better IATV model each time.

## 5. Conclusions

This show-and-tell demonstrates one possible implementation of the IATV research task, as the stand-alone IATV software Tungnaá, and describes the process of creating a bespoke IATV dataset and model within the performance Bla Blavatar vs. Jaap Blonk. By describing the IATV task and its requirements, we hope to spark more artist-led work in the field of voice technology research. By presenting an example implementation of software that meets the IATV requirements and has been tested in a demanding live performance situation, we demonstrate the feasibility and potential of IATV systems, providing tangible evidence of the concept’s viability.

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